He's a real weasel: Processing adjectival markers of epistemic stance and deriving non-literal meaning

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Abstract

We investigate whether the English adjective *real* can serve as a predictive cue in the processing of metaphors, as measured via self-paced reading and eye-tracking while reading. Crucially, alongside its conceptual meaning of reality, *real* functions as a marker of epistemic stance with an intensificational meaning, which may give rise to ambiguities in the online processing of non-literal utterances. We test whether combining *real* with a noun used metaphorically (e.g. *He is a real weasel*) cues a non-literal interpretation as strongly as constructions designed to be more compatible with a metaphoric interpretation (e.g. *He is a devious weasel*) and more strongly than constructions designed to be less compatible with a metaphoric interpretation (e.g. *He is a furry weasel*). Our results show that constructions with *real* cue a non-literal interpretation as strongly as constructions designed to be interpreted more metaphorically, and more strongly than constructions designed to be interpreted less metaphorically. The results also suggest that both the adjective's intensificational and conceptual meaning are activated, and that this does not hinder the derivation of a non-literal interpretation. We discuss these results in light of work on utterance interpretation, highlighting the role stance markers might play in the derivation of pragmatic meaning.

Keywords: epistemic stance, stance marker, metaphor, pragmatic processing, predictive processing, incremental interpretation

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1. Introduction

Adjectives provide language users with means of categorizing the world around them according to properties related, for instance, to psychological or social traits (e.g. *devious*, *brave*) or perceptual ones (e.g. *furry*, *tall*). While prototypically used to describe some inherent or perceived property attributed to a referent (e.g. The Dutch are *tall* people), adjectives can also be used to highlight a noun's scalar degree, such as when describing a situation as a *real* mess, where *real* does not necessarily signal the 'reality' of the mess but rather its intense or particularly 'messy' character. Crucially, in such a usage the adjective can be said to mark a language user's epistemic stance on the messiness of the situation, not contributing truth-conditional information to the utterance it appears in but rather adding a layer of subjectivity to it. In the present study, we investigate whether the English adjective *real* can be used as a pragmatic cue in the incremental processing of non-literal utterances such as *He is a real weasel sometimes during games*, as measured in a word-by-word self-paced reading task as well as an eye-tracking while reading task. Before introducing the experiments and reporting our findings, we discuss how an adjective like *real* might be interpreted vis-à-vis its polysemous/ polyfunctional character as both a marker of truth and of intensification.

1.1. Stance markers as cues to utterance interpretation

Plenty of psycholinguistic research focused on pragmatic processing has looked at the interpretation of utterances where the usage of an expression like the adjective *tall* gives rise to an inference such that, given multiple possible utterance continuations (e.g.*Pick up the tall candle* vs. *Pick up the tall glass*), interpreters quickly converge on a likely continuation before processing subsequent linguistic material (e.g. Sedivy et al., 1999; Gardner et al., 2021; Grodner and Sedivy, 2011; Rubio-Fernandez and Jara-Ettinger, 2020). Key to these so-called contrastive inferences is the fact that the adjective can lead interpreters to anticipate subsequent linguistic material in the utterance on the basis of pragmatic expectations to do with what an informative utterance continuation might be (e.g. *Pick up the tall candle* is more informative than *Pick up the tall glass* in a context where both a tall and a short candle are possible referents but only one glass). Whatever the actual mechanism involved in generating a relevant pragmatic contrast between competing utterance continuations, the underlying assumption is

that one is able to anticipate what the content of the utterance might be given that the adjective constrains the most likely truth-conditional meaning associated with the unfolding utterance.

However, not all linguistic forms constrain the truth conditions associated with an utterance. Take, for instance, *damn* in *The damn dog is on the couch*. Here, the adjective signals that the producer of the utterance has a negative stance or attitude towards the dog sitting on the couch. Importantly, while *damn* does not change the truth conditions associated with the utterance *The dog is on the couch*, it serves as an ostensive cue that the producer is not happy with the fact that the dog is sitting on the couch, and it may impact how the utterance is interpreted incrementally. Indeed, it has been found that comprehenders are able to anticipate content in discourse contexts where there is a negative speaker attitude toward said content based on similar adjectives in Italian (Ronderos and Domaneschi, 2023). While the actual inferential status of these so-called conventional implicatures (Potts, 2007, 2003, 2012) is still debated in the literature (see Asudeh and Giorgolo, 2020 for a discussion), key to the underlying interpretive process is the fact that processing the pragmatically relevant lexical item does not result in a change to the truth-conditional meaning of the processed utterance but rather in the derivation of an altogether different layer of meaning.

Even though the adjective *damn* does not encode anything related to a property of the noun it modifies it nevertheless signals a heightened negative emotion or attitude towards a proposition associated with its host utterance. In fact, the meaning of the adjective remains unaltered regardless of its position within the utterance (e.g. *The damn dog is on the couch* $\approx$ *The dog is on the damn couch*), affecting the utterance as a whole and not just the noun it modifies. Not all lexical items which mark a producer's stance on a proposition behave as straightforwardly as *damn*, however. In fact, some markers of stance preserve a conceptual meaning alongside their meaning as stance markers. One example is that of epistemic adjectives like *actual*, *literal*, and *real* in English. While these adjectives can be used in their lexical sense to describe the authenticity of something (e.g. *This is a real watch*, meaning that it is not a fake one), they can also be used as markers of intensification, as in *He is a real liar*. In saying that someone is a *real* liar a producer marks the degree to which they commit to the truth or validity of the information being conveyed by the noun *liar* (Conrad and Biber, 2000). More specifically, in such a construction *real* signals that the person at hand is a liar to an extent which is above some relevant threshold of liar-ness, a

threshold which is ultimately subjective. In short, *real* is used to intensify the relevant semantic property associated with the noun *liar*.

Crucially, while in their intensifying usage adjectives like *actual*, *literal*, and *real* impart a subjective assessment to the meaning of the noun they intensify, thus serving as markers of subjective stance, they can still be said to mark the truth of the proposition associated with the utterance they occur in, thus preserving traces of their conceptual epistemic meaning. In other words, at face value, the conceptual meaning associated with the lexical item *real* seems to be more salient than its intensificational meaning such that the two are not easily distinguished. *Real*'s adverbial counterpart *really* provides an interesting example of how an intensifier's stance meaning can, however, become more salient than its conceptual meaning depending on how it is used (Paradis, 2002). Indeed, much like *real*, *really* can also be used as a marker of intensification; crucially, the different meaning profiles associated with it correlate with particular syntactic configurations, such that the closer the adverb is to either syntactic periphery the more of an objective, and thus conceptual, meaning it tends to have. For example, in 1 the adverb signals a subjective emphasis on the truth of the proposition at hand, which relates to its meaning of 'reality', while in 2 the same adverb signals a reinforcement of the scalar degree of *sneaky*, which only indirectly relates to any objective truth of the predicate *He is sneaky*.

1. *He really is sneaky* $\sim$ *I commit to the fact that he is sneaky*
2. *He is really sneaky* $\sim$ *He is sneaky above some relevant threshold of sneaki-ness*

While both usages of *really* show an epistemic flavor to them, in its emphasizing usage in 1 the relation between the lexical content of 'reality' and the subjective attitude bound to the verb is much more transparent compared to the intensifying usage in 2, where the relation between the lexical content of 'reality' and the intensification of *sneaky* is less transparent. Unlike its adverbial counterpart, *real* mostly occurs right before the head it modifies, meaning that its syntactic position, even when it is being used as an intensifier, provides no reliable cue to as to whether it is to be interpreted in terms of its meaning of intensification or in terms of its conceptual meaning. That is, while *really* in *He is really sneaky* seems to rather straightforwardly intensify *sneaky*, without clear traces of any epistemic meaning related to 'reality' (in the sense of literal truth), *real* in *He is real sneaky* might be said not

only to indicate an intensified degree of *sneaky* but also to impart some sense of 'reality' to it, such that its conventionalized meaning of intensification is not clearly distinguishable from its conceptual meaning (unlike *fake* in *He is fake sneaky*). How does this multidimensionality in meaning impact the interpretation of utterances with intensifying adjectives? In other words, is the epistemic meaning of an adjective like *real* more salient than its intensificational meaning or are interpreters able to distinguish between the two on the fly despite a potential ambiguity?

In the present study, we investigate precisely that question, namely whether intensifying adjectives, as markers of subjective epistemic stance which do not affect the truth conditions associated with an utterance, modulate *incremental* interpretation in online processing. Crucially, we are interested in whether such adjectives can be used as pragmatic cues in the incremental processing of nominal metaphors, such that they cue a likely non-literal interpretation in spite of their conceptual meaning related to truth. While adjectives are known to act as predictive cues to interpretation in the online processing of language (e.g. Sedivy et al., 1999; Gardner et al., 2021; Grodner and Sedivy, 2011; Rubio-Fernandez and Jara-Ettinger, 2020), most psycholinguistic studies to date have investigated whether adjectives provide cues to the literal meaning of an unfolding utterance, usually in the form of information related to color (e.g. *blue*), material (e.g. *glass*), or some other physical property (e.g. *tall*). Our goal here is to test whether comprehenders process non-literal, metaphoric statements predictively when an adjective in the predicate of the construction provides a cue to the producer's stance on the content of the utterance, as opposed to semantic information pertaining to the content itself. Unlike previous work looking at the processing of adjective-modified noun phrases, we focus our attention on the processing of adjectives which function as markers of intensification despite having a lexical semantics related to truth or literality. What this means is that, unlike in the case of adjectives which contribute information which directly affects the truth-conditional meaning of an utterance, adjectives with a meaning of intensification might render an utterance temporarily ambiguous not because they are denotationally underspecified but rather because their intensificational and conceptual meanings might induce different, if not conflicting, interpretations during processing.

Based on the observation that there might be a non-trivial interplay between the intensificational and conceptual meaning of epistemic adjectives, such adjectives are expected to affect incremental interpretation differentially

depending on whether they cue an interpretation more or less congruent with the most likely interpretation in context, whatever that may be. Crucially, in a metaphoric statement like *He is a real weasel sometimes during games*, where *weasel* refers to a person who is sneaky or devious, the adjective *real* can be said to signal both a non-literal as well as a literal meaning. At face value, this might suggest a potential (temporary) ambiguity in interpreting the utterance, at least until a cue like *sometimes*, which reinforces the likely non-literal interpretation of *He is a real weasel*, enters the frame and further constrains the interpretation process. However, this *prima facie* ambiguity is not expected under the assumption that comprehension involves dynamically weighing the cues available in a signal at any given moment in processing and deriving therefrom the most likely interpretation given general discourse expectations (Degen and Tanenhaus, 2019; Rohde and Kurumada, 2018, see also MacDonald, 1994; McRae et al., 1998), including the expectation that all cues in the signal are relevant given the producer's pragmatic goals. Indeed, under such an expectation-based account of (incremental) interpretation, the meaning of intensification of *real* in the case above is arguably more congruent with a non-literal interpretation of *weasel* than its conceptual meaning of 'reality', which therefore suggests no qualitative ambiguity in interpreting the utterance, assuming that the former takes precedence over the latter.

In order to test the hypothesis that a potentially ambiguous adjective like *real* can provide a cue to non-literality in a metaphoric utterance, we investigate how readers process metaphoric sentences containing either a stance adjective or an adjective whose conceptual meaning acts as a more obvious cue to non-literality. In Experiment 1, a self-paced reading experiment, we focus on how readers process a cue to non-literality which appears after the adjective-modified metaphoric noun. In Experiment 2, an eye-tracking while reading replication of Exp. 1, we focus on how readers process both the post-noun critical region as well as the preceding adjective and the noun itself. In Experiment 3, a revised and extended version of Exp. 2, we focus on how readers process both literal and non-literal sentences containing stance adjectives.

2. Experiment 1 - Self-paced reading

The experiment was pre-registered under `https://osf.io/vdjc4/`. Ethics approval was obtained from the Ethics Review Board at the University of Edinburgh.

2.1. Method

2.1.1. Participants

80 self-reported monolingual speakers of English were recruited using the crowdsourcing platform Prolific. Participants received £2.18 for their participation in the experiment, which lasted on average no longer than 15 minutes. Data collection was conducted entirely online and all collected data was stored on EU servers.

2.1.2. Materials and design

Participants were asked to read 33 English sentences. Their task was to read each sentence for comprehension, according to the procedure described below. There were 15 critical trials containing nominal metaphors, as in the examples 1-3 below, as well as 18 filler trials, 12 of which also contained adjective-modified nominal metaphors (e.g. *He is the liveliest fish in the tropical aquarium*) and 6 of which contained literal, non-metaphoric statements (e.g. *This afternoon we should take a drive in the countryside*).

All critical statements consisted of nominal metaphors, such that the topic always consisted of a third-person singular pronoun (either *he* or *she*) and the vehicle, i.e., the noun in the predicate of the construction, always consisted of an animal conventionally associated with a psychological or physical attribute characteristic of people (e.g. *weasel* $\sim$ a person who is sneaky or devious). The critical sentences always contained the adverb 'sometimes' after the noun, the adverb functioning as a strong cue to non-literality. There were 15 critical animal nouns (see OSF for the item list), meaning that, on each trial, participants were presented with an altogether different metaphor. Crucially, each critical noun appeared in one of three experimental conditions, as in the triplet below:

Metaphor-supporting He is a *devious* weasel sometimes during games.

Stance-marking He is a(n) *actual/ literal/ real* weasel sometimes during games.

Literal-supporting He is a *furry* weasel sometimes during games.

The only difference between the sentences above is the lexical item appearing directly before the vehicle in the metaphor. Crucially, by design, constructions in the *Metaphor-supporting* condition are meant to cue an interpretation which is more likely metaphoric, i.e., one where a sneaky or

devious person is being referred to (e.g. He is a *devious weasel*); constructions in the *Literal-supporting* condition, on the other hand, are meant to cue an interpretation which is less likely metaphoric, i.e., one where what is being referred to is a potentially an animal and not a person (e.g. He is a *furry weasel*). As discussed in the introduction, in a construction like *He is a real weasel* the adjective *real* can function as an intensifier of the noun *weasel*. Items in the *Stance-marking* condition contained one of three intensifying adjectives, either *actual*, *literal*, or *real* (counterbalanced across participants). Note that in the *Metaphor-supporting* and *Literal-supporting* conditions the lexical items preceding the metaphoric nouns were often attributive nouns, as in *She is a mother hen* or *He is a mountain lion*, making the pair a lexical phrase. Crucially, aside from the critical noun phrase, all critical items also contained a modifier adverbial phrase headed by *sometimes*. By design, *sometimes* was meant to cue a likely metaphoric interpretation of the sentence, which, from an incremental perspective, either supports or clashes with the unfolding interpretation, depending on whether that interpretation is either more likely metaphoric, as should be the case in the *Metaphor-supporting* items (i.e. *He is a devious weasel ...*), or less likely metaphoric, as should be the case in the *Literal-supporting* items (i.e. *He is a furry weasel ...*). In the case of *Literal-supporting* items, while not fundamentally changing the underlying meaning of the sentence, *sometimes* may serve as a cue that points readers toward a non-literal meaning. Items were counterbalanced so that participants only saw each animal in a single experimental condition.

2.1.3. Procedure

Written instructions were provided prior to the actual task, followed by three practice trials which mimicked the exact procedure of the test trials. On each trial, participants were presented with a target sentence, as described in the section above. The sentence was masked at the beginning of the trial, underscores marking the location of each individual word. Above the underscores was the written cue "Press the SPACE bar to reveal the words". Participants were instructed to press the space bar in order to reveal each word in the sentence, hence they read the sentence one word at a time, in a self-paced manner. Once they read the last region in the sentence, pressing the space key would trigger a button labeled "Next", which had to be clicked in order to advance to the next trial. Once every third trial, instead of the "Next" button, participants were presented with a comprehension question

relating to the sentence they read. They were asked to choose between two response options and received written feedback regarding their choice, which had the sole purpose of keeping participants engaged with the task of reading for comprehension.

2.2. Hypotheses

The hypothesis entertained here is that the lexical item preceding the metaphoric noun should affect incremental sentence interpretation differentially depending on whether its semantics cues an interpretation more or less congruent with a non-literal, metaphoric meaning. Importantly, all predictions alike relate to the processing signatures measured at the adverb *sometimes*, which is the region directly following the noun in the critical sentence frame. This entails an effect such that sentences with a *Literal-supporting* lexical item (*He is a furry weasel sometimes ...*) are expected to be harder to process at the adverb *sometimes* compared to sentences with a *Metaphor-supporting* lexical item (*He is a devious weasel sometimes ...*), given that the adverb provides a cue which potentially conflicts with a literal interpretation of *He is a weasel*. To be explicit, in the *Literal-supporting* condition, *sometimes* may be taken to indicate an anomalous reading, hence reading times at this point should be increased, relative to other conditions. In turn, if the *Metaphor-supporting* condition is interpreted without anomaly as a figurative description, reading times at *sometimes* should be reduced, relative to the anomalous (literal) reading. Crucially, for our main question, the actual interpretation of the *Metaphor-supporting* and *Literal-supporting* conditions is not relevant (it is just relevant that we expect them to differ), and in any case we are unable to verify exactly how readers interpret sentences using the methodology adopted here. More importantly, we are interested in how conditions with *real* pattern: if they pattern more like the *Literal-supporting* items, we assume that this indicates that *real* was interpreted more according to its semantics related to truthfulness or reality, whereas if they pattern more like the *Metaphor-supporting* items, we assume that *real* was interpreted as a cue to non-literal interpretation, i.e. in terms of its conventionalized meaning as an intensifier.

2.3. Results

The data and analysis scripts are available for inspection under `https://osf.io/u6f98/`.

2.3.1. Data cleaning

Prior to any analysis, the data set was cleaned according to criteria defined in the pre-registered analysis plan, which include both participant-based and response-based exclusion criteria. We first excluded the data from any participant who self-reported being a monolingual speaker of a language other than English (n=1). Any participant who did not reach an accuracy of 80% on the comprehension questions would be removed. All participants exceeded this threshold. Finally, any participant who deviated in more than 30% of their trials in more than 2.5 standard deviations from the grand mean for a given condition, as measured at any individual sentence region, would be removed. No participant exceeded this threshold. Altogether, cleaning resulted in a data set which included data from 79 participants.

In addition to excluding one participant as per the criteria above, we excluded any individual data point which deviated by more than 2.5 standard deviations from the grand mean for a given condition (1% of the data). Finally, we removed any data point lying outside of a pre-established range of interest set between 100 and 1000 ms (2% of the data). This reflects a meaningful range for the reading of individual words in self-paced reading.

2.3.2. Confirmatory analyses

Figure 1 shows the mean reading times, aggregated over all experimental items and participants.

The figure shows that, at the adverb *sometimes*, there is a difference between the *Literal-supporting* condition and both the *Metaphor-supporting* and *Stance-marking* conditions, which only differ negligibly from one another. This suggests that comprehenders' online expectations vary as a function of incrementally combining semantic information from the noun and the word preceding it.

In order to quantify uncertainty over the results reported below, we fitted a Bayesian hierarchical model predicting log-transformed RTs at the critical region *sometimes* as a function of the experimental conditions, with the condition *Literal-supporting* dummy-coded as the reference level. The model included random slopes and intercepts for both participants and items. We model group-level variation within the items nesting them within the animal nouns, given that random variation at the level of the sentences should be accounted for both in terms of the critical mappings between the nouns and the word immediately preceding them as well as in terms of any idiosyncrasies tied to the individual animals used as vehicles in the metaphors. The

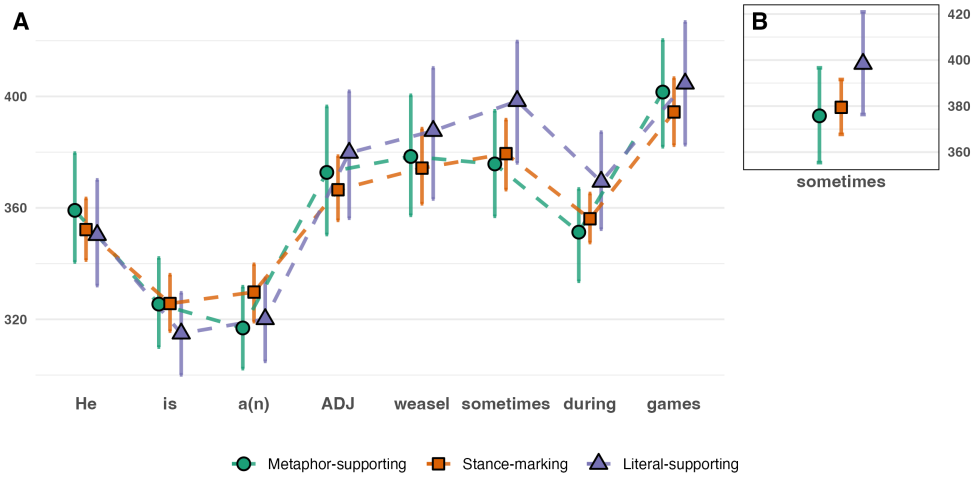


Figure 1: Panel A shows the mean reading times at each sentence region. Panel B shows a close-up of the mean reading times at the adverb region. *Metaphor-supporting* items are shown in green, *Stance-marking* in orange, and *Literal-supporting* in purple. The error bars represent 95% bootstrapped confidence intervals.

model, fitted using the *R* package `brms` (Bürkner, 2017), had the following form, shown in `brms` syntax:

$$\begin{aligned} \log(\text{RT}) \sim \text{condition} + \\ (1 + \text{condition} \mid \text{participant}) + \\ (1 + \text{condition} \mid \text{animal/ item}) \end{aligned}$$

For each highlighted result, we report the evidence provided by the statistical model in favor of the empirically attested differences (or lack thereof). In a Bayesian statistical framework, one is interested in the joint posterior distribution of the parameters of the model, which indicates a plausible range of values for the parameters given the model and the data at hand. We report a credibility interval (CrI) and the posterior probability that the parameter of interest, β, is larger than zero ($P(\beta > 0)$). While this measure of evidence is gradient, one can speak of strong evidence for an effect when zero is not included in the CrI and $P(\beta > 0)$ is close to either zero or one, whereas when $P(\beta > 0)$ is around 0.5 one can speak of strong evidence for the absence of any effect. Concretely, we are interested in the difference between estimated mean RTs.

Visual inspection of the results in Figure 1 suggests that, at the *sometimes* region, items in the *Metaphor-supporting* condition are read faster than items in the *Literal-supporting* condition, in line with the baseline prediction. This result finds relatively strong quantitative support in the respective statistical model ($RT_{literal}$ - $RT_{metaphor}$, β = 16 ms, 95% CrI [-4.32, 36.76], $P(\beta > 0) = 0.93$). As for the items in the *Stance-marking* condition, the descriptive results suggest that, much like items in the *Metaphor-supporting* condition, they are read faster than items in the *Literal-supporting* condition, in line with the prediction derived from the hypothesis of stance markers as predictive pragmatic cues. This result finds particularly strong quantitative support in the respective statistical model ($RT_{literal}$ - RT_{stance}, β = 15 ms, 95% CrI [-3.16, 30.96], $P(\beta > 0) = 0.95$). The descriptive results further suggest no reliable difference between the items in the *Stance-marking* and *Metaphor-supporting* conditions ($RT_{metaphor}$ - RT_{stance}, β = -1 ms, 95% CrI [-20.31, 15.76], $P(\beta > 0) = 0.45$).

Overall, then, in Exp. 1, the intensifying adjectives *real*, *actual* and *literal* seemed to act as cues toward non-literal meaning (or, at the very least, are not interpreted in ways that render the sentence anomalous), supporting the interpretation of sentences as (potentially) metaphorical rather than literal.

3. Experiment 2 - Eye-tracking while reading (replication Exp. 1)

The aim of Experiment 2 was to replicate Exp. 1 using eye-tracking while reading, which provides a more sensitive measure of incremental processing than self-paced reading. Crucially, it also captures re-reading behavior (regressions), which may give additional information about how processing difficulties are resolved during natural reading. Ethics approval was obtained from the Ethics Review Board at the University of Birmingham.

3.1. Method

3.1.1. Participants

53 monolingual speakers of English, none of whom reported dyslexia or any other reading conditions, were recruited from the student population of the Department of Linguistics and Communication of the University of Birmingham, United Kingdom. Participants received course credit in exchange for their participation in the experiment, as part of a lab session incorporating a

range of studies. Data collection was conducted in a dedicated laboratory facility at the Department of Linguistics and Communication and all collected data was stored on university servers.

3.1.2. Materials and design

The materials were the same as in Exp. 1, meaning that each critical item appeared in one of three experimental conditions, as in the triplet below:

Metaphor-supporting He is a *devious* weasel sometimes during games.

Stance-marking He is a *real* weasel sometimes during games.

Literal-supporting He is a *furry* weasel sometimes during games.

We decided to only include one stance adjective, namely *real*, in the *Stance-marking* condition, given that there were negligible differences between the adjectives in Exp. 1 and in order to simplify the design. Items were distributed across three counterbalanced lists. The same set of fillers as in Exp. 1 were supplemented with an additional set of similarly structured sentences to give a total of 40 distractors, used across all three lists.

3.1.3. Procedure

The experiment was built in the Experiment Builder software (version 2.3.38) and utilized an Eyelink 1000+ desk-mounted eye-tracker (SR Research), tracking monocularly (left eye) at 500 Hz. Participants were seated at approx. 60 cm from a 1920x1080 screen with a refresh rate of 60 Hz. Oral and written instructions were provided prior to the actual task, followed by a standard 9 point calibration and validation procedure.

Recalibrations were performed as required during the study to maintain tracking accuracy. Prior to each trial a drift correct dot appeared at the top of the screen to allow for drift-checking, then participants were presented with a target sentence, as described in the section above. Sentences appeared across the middle of the screen in 18pt Courier New font. Participants were asked to read the sentence for comprehension, then press the space bar once they had finished. On selected trials, participants were presented with a comprehension question relating to the sentence they read. They were asked to use the mouse to choose one of three response options, which had the sole purpose of keeping participants engaged with the task of reading for

comprehension. Comprehension questions were asked after all *Metaphor-supporting* and *Stance-marking* items, and after 19 of the filler items (just over one third of all trials overall). We did not ask comprehension questions after *Literal-supporting* items as the deliberate incongruency they included might make people more likely to answer incorrectly. Five practice items were presented, followed by the 15 critical items and 40 fillers presented in random order. Average completion time was around 10 minutes.

3.2. Hypotheses

The hypothesis entertained here was the same as in Exp. 1, namely that stance adjectives act as strong cues to non-literality in the processing of metaphoric statements. We pre-registered a series of eye-tracking analyses pertaining to different sentence regions and involving different reading measures. All analyses are described in detail in sections 3.3.2 and 3.3.3 below.

3.3. Results

The data and analysis scripts are available for inspection under `https://osf.io/u6f98/`. Unless otherwise noted, all analytic choices and procedures reported below follow the pre-registered analysis plan. Data was pre-processed using Eyelink DataViewer software (version 4.4.1).

3.3.1. Data cleaning

Prior to any analysis, the data was checked visually for poor calibration or track loss. The eye tracker was found to be poorly calibrated whenever visual inspection of the data led to a high number of trials being removed, or whenever data consistently required vertical alignment that was not straightforward. Participants who consistently had poor calibration or who failed to follow instructions (e.g. not pressing the spacebar after reading each sentence) were removed, leaving usable data from 41 participants. Individual trials where obvious track loss occurred were removed, as were trials where both the adjective and noun regions were skipped, or trials where *sometimes* (the critical region) received no fixations at all. Overall, this led to the removal of 5.2% of usable data and left 583 trials for analysis. Since all reading took place on a single line, data was also vertically aligned to correct any minor deviations in terms of the y-position. The data was then cleaned according to the four-stage process in the Eyelink DataViewer software (which

first combines short proximal fixations, then removes extreme values, i.e. fixations below 100ms, and over 800ms).

According to our pre-registered plans, any participant who did not reach an accuracy of 80% on the comprehension questions would be removed. All participants exceeded this threshold (mean = 91.7%, SD = 27.6).

3.3.2. Reading times

First, we report the mean reading times across sentence regions, as shown in Figure 2 below. The figure includes three different reading measures, first pass reading times, second pass reading times, and total reading times. All relevant results are summarized in Table 1 below. Lots of different measures are available in eye-tracking research, reflecting different aspects of text processing. In general terms, early measures such as first pass reading are taken to reflect immediate lexical access and retrieval, whilst later measures, such as second-pass / total reading time, are assumed to more reflect overall meaning integration across the whole text (see e.g. Staub and Rayner 2007; Conklin et al. 2018 for full discussion of ET measures). Importantly, measures are not independent here: total reading time includes first pass reading time and any subsequent re-reading. For this reason, second pass reading time was included as a way to separate out early / late effects.

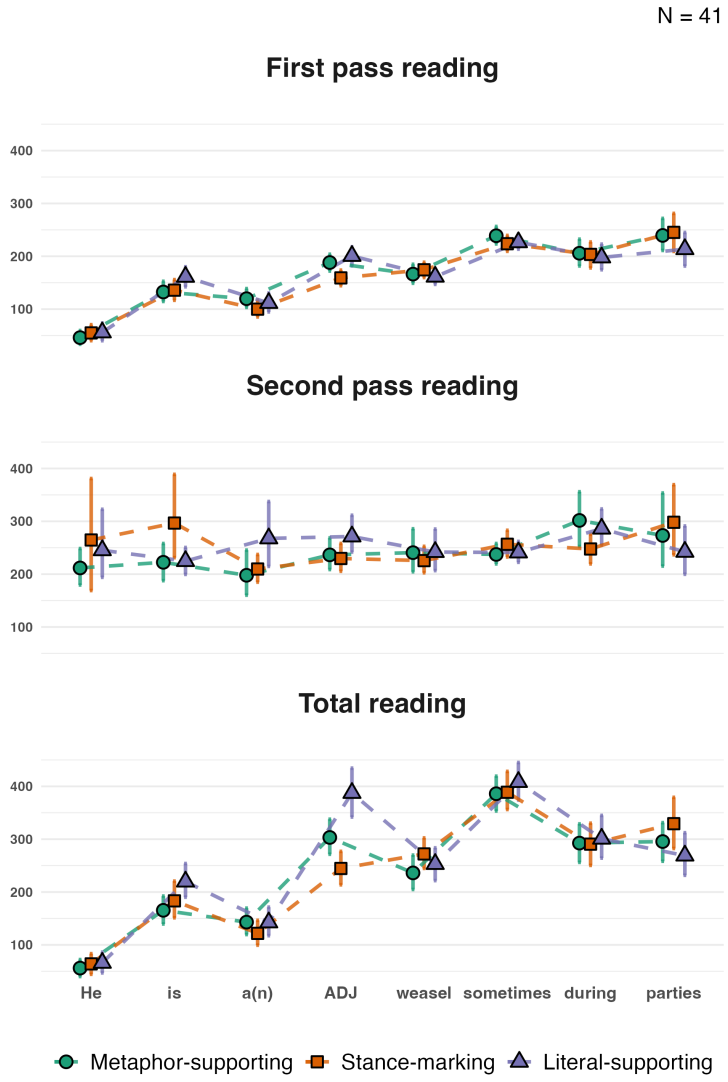


Figure 2: Mean reading times (y-axis) at each sentence region (x-axis). The top panel shows first pass reading times, the middle one second pass reading times, and the bottom one total reading times. *Metaphor-supporting* items are shown in green, *Stance-marking* in orange, and *Literal-supporting* in purple. The error bars represent 95% bootstrapped confidence intervals. Note that second pass reading times do not include any zero values.

		Metaphor-supporting	Stance-marking	Literal-supporting
adjective	First pass	188 (110)	159 (105)	201 (123)
	Regression path	297 (261)	247 (248)	331 (279)
	Second pass⋆	80 (135)	64 (116)	128 (176)
	Second pass*	237 (128)	230 (103)	271 (164)
	Total	303 (242)	245 (218)	387 (333)
	Reg. IN	0.35 (0.48)	0.29 (0.45)	0.42 (0.49)
	Reg. OUT	0.20 (0.40)	0.15 (0.36)	0.23 (0.43)
noun	First pass	166 (124)	174 (108)	161 (107)
	Regression path	244 (273)	257 (265)	243 (285)
	Second pass⋆	54 (120)	76 (124)	69 (138)
	Second pass*	241 (138)	226 (107)	242 (158)
	Total	236 (224)	272 (213)	253 (232)
	Reg. IN	0.24 (0.43)	0.29 (0.45)	0.23 (0.42)
	Reg. OUT	0.14 (0.35)	0.15 (0.36)	0.12 (0.33)
sometimes	First pass	239 (119)	224 (102)	226 (103)
	Regression path	354 (304)	312 (255)	326 (302)
	Second pass⋆	114 (144)	114 (153)	127 (141)
	Second pass*	237 (92)	257 (126)	241 (102)
	Total	386 (221)	389 (246)	408 (255)
	Reg. IN	0.36 (0.48)	0.33 (0.47)	0.37(0.48)
	Reg. OUT	0.16 (0.37)	0.14 (0.36)	0.13 (0.34)
post-critical	First pass	485 (330)	470 (303)	447 (317)
	Regression path	964 (610)	1057 (883)	1077 (854)
	Second pass⋆	132 (158)	166 (326)	136 (224)
	Second pass*	443 (300)	475 (403)	361 (229)
	Total	637 (376)	673 (485)	616 (229)
	Reg. IN	0 (0)	0 (0)	0 (0)
	Reg. OUT	0.60 (0.49)	0.58 (0.49)	0.64 (0.48)

Table 1: Reading measures at the adjective, noun, *sometimes*, and collapsed post-critical regions. Reading times are shown in milliseconds, regressions in percentages (with standard deviations within parentheses). ⋆ = Second pass reading times including zeroes (i.e. where no re-reading took place); * = Second pass reading times not including zero values (i.e. only re-reads).

We pre-registered confirmatory analyses focused on the critical region *sometimes* which included first pass reading times and regression path duration (see `https://osf.io/adrjx/`). As in Exp. 1, our baseline expectation was that *Metaphor-supporting* items would be read faster than *Literal-supporting* items both in first pass reading times and regression path duration, given the congruency between non-literal cues in the former but not in the latter. Crucially, in line with this expectation, we predicted that *Stance-marking* items would be read as fast as *Metaphor-supporting* items and faster than *Literal-supporting* items both in first pass reading times and regression

path duration, under the hypothesis that stance adjectives act as strong cues to non-literality. Figure 2 shows that, at *sometimes*, differences between conditions in first pass reading times are negligible, unlike what was expected by our baseline expectation and by the hypothesis of stance adjectives as strong cues to non-literality. Table 1 shows that the same is true of regression path duration, except when comparing *Stance-marking* and *Metaphor-supporting* items, the former being read faster than the latter.

While *sometimes* was our region of interest as per the original design of Exp. 1, we also pre-registered exploratory analyses focused on both pre- and post-critical sentence regions. The pre-critical adjective and noun regions introduce the relevant semantic information which, as per the design, was expected to make *sometimes* more or less predictable depending on the experimental manipulation. We pre-registered exploratory analyses focused on the pre-critical regions which included first pass reading times, regression path duration, and total reading times. While second pass reading time was not part of our pre-registered analyses, we decided to include it as, despite being similar to total reading times, it does not take first pass into account, meaning that it only involves non-cumulative reading measurements and, as such, might be considered as a more informative index of late processing. In what follows, we include second pass reading times where in our registration only total reading times were mentioned.

Figure 2 shows that, at the adjective, *Metaphor-supporting* items are read descriptively faster than *Literal-supporting* items in second pass reading times (difference = 34 ms). According to Table 1, the same is true for regression path duration (34 ms). The difference between *Metaphor-supporting* and *Literal-supporting* items is negligible in first pass. Crucially, *Stance-marking* items are also read faster than *Literal-supporting* items, both in first and second pass reading times and regression path duration. The difference is numerically similar in first and second pass (first pass = 42 ms; second pass = 41 ms) and larger in regression path (84 ms). The differences in second pass reading times, also apparent in total reading times, suggest that the adjective biases readers toward a non-literal interpretation more strongly in *Metaphor-supporting* and *Stance-marking* items compared to *Literal-supporting* items. However, this only becomes evident once readers have encountered further cues downstream in the sentence, i.e. once *sometimes* as a further cue to non-literal meaning makes the *Literal-supporting* condition less congruent than its *Metaphor-supporting* and *Stance-marking* counterparts. At the noun, reading time differences between conditions are negligible in first and second

pass and regression path.

We also pre-registered exploratory analyses of the post-critical regions, which we collapsed together, including first pass reading times and total reading times. According to Table 1, differences between conditions are negligible, both in first pass and total reading times. Given the attested differences from Fig. 2 and Table 1, we focus on the second pass reading times at the adjective (ADJ). In order to quantify uncertainty over the results reported below, we fitted a Bayesian hierarchical model predicting log-transformed reading times at the adjective region as a function of the experimental conditions, with the condition *Literal-supporting* dummy-coded as the reference level. Given that readers may not necessarily read a word a second time after first pass, we excluded trials on which second pass reading times were 0 (64% of the data at the adjective, see Second pass* in Table 1). In order to account for the light right skew in the response distribution, we fitted a model using a skewed normal distribution. In addition to a location (μ) and scale (σ) parameter, just like in a normal linear model, this model also estimates a skew parameter (α). The model included random slopes and intercepts for both participants and items. We model group-level variation within the items nesting them within the animal nouns. The model, fitted using the *R* package `brms` (Bürkner, 2017), had the following form, shown in `brms` syntax:

$$\begin{aligned} \log(\text{reading time}) \sim \text{condition} + \\ (1 + \text{condition} \mid \text{participant}) + \\ (1 + \text{condition} \mid \text{animal/ item}) \end{aligned}$$

We report estimated differences in RTs as well as 95% credibility intervals (95% CrI [lower bound, upper bound]) around those estimates. We also report the posterior probability that the difference in estimated means between any two conditions is larger than zero. Concretely, we are interested in the difference between estimated means of *Metaphor-supporting* and *Literal-supporting* items ($\text{RT}_{\text{literal}}$ - $\text{RT}_{\text{metaphor}}$), *Stance-marking* and *Literal-supporting* items ($\text{RT}_{\text{literal}}$ - $\text{RT}_{\text{stance}}$), and *Stance-marking* and *Metaphor-supporting* items ($\text{RT}_{\text{metaphor}}$ - $\text{RT}_{\text{stance}}$). We plot the differences as predicted by the model in Figure 3.

Visual inspection of the results in Fig. 2 suggests that, at the adjective region, *Metaphor-supporting* items have shorter second pass reading times than *Literal-supporting* items in second pass. This result finds relatively strong quantitative support in the respective statistical model ($\text{RT}_{\text{literal}}$ -

$RT_{metaphor}$ = 27 ms, 95% CrI [-8.4, 57.3], $P(RT_{literal} > RT_{metaphor}) = 0.94$). As for *Stance-marking* items, the descriptive results suggest that, much like *Metaphor-supporting* items, they have shorter second pass reading times than *Literal-supporting* items. This result also finds somewhat strong quantitative support in the respective statistical model ($RT_{literal}$ - RT_{stance} = 26 ms, 95% CrI [-9.5, 57.9], $P(RT_{literal} > RT_{stance}) = 0.94$). The descriptive results further suggest no reliable difference between the items in the *Stance-marking* and *Metaphor-supporting* conditions, a result which is supported by the model's posterior predictions ($RT_{metaphor}$ - RT_{stance} = -1 ms, 95% CrI [-36.5, 43.0], $P(RT_{metaphor} > RT_{stance}) = 0.48$).

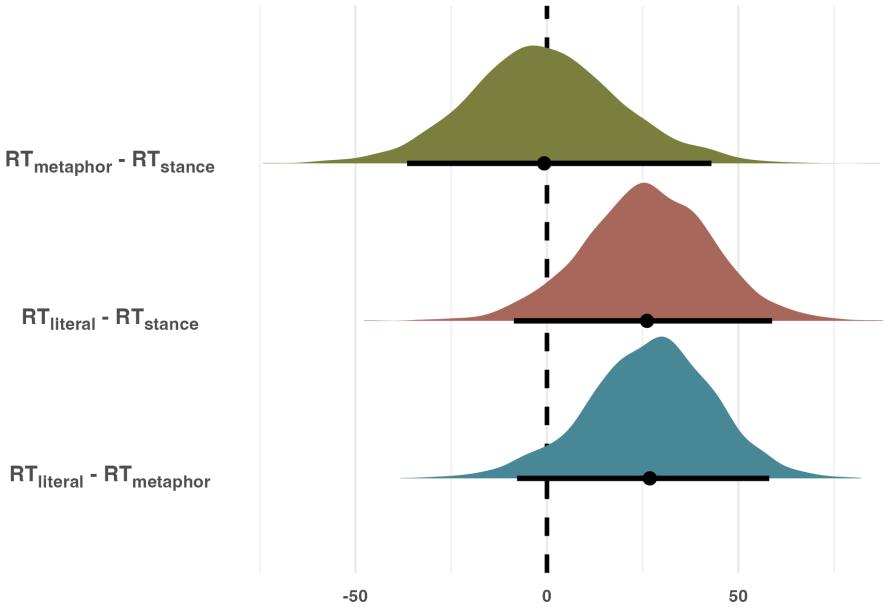


Figure 3: Differences in estimated second pass reading times at the adjective (x-axis). The difference between *Literal-supporting* and *Metaphor-supporting* items ($RT_{literal}$ - $RT_{metaphor}$) is shown in blue, the one between *Literal-supporting* and *Stance-marking* ($RT_{literal}$ - RT_{stance}) in red, and the one between *Metaphor-supporting* and *Stance-marking* ($RT_{metaphor}$ - RT_{stance}) in green. The black circles represent the mean estimated differences and the horizontal bars represent the respective 95% credibility intervals.

3.3.3. Regression rates

We pre-registered a confirmatory analysis focused on regressions out of the *sometimes* region (see `https://osf.io/adrjx/`). Figure 4 below shows, on the left-hand side, the mean proportion of regressions out of *sometimes*.



Figure 4: Mean proportions of regressed trials (y-axis). Mean proportions of regressed trials (fully colored shapes), overlaid to participant means (shaded shapes with varying sizes). Values closer to 1 on the y-axis represent a larger proportion of trials on which participants regressed out of *sometimes* (left sub-figure) and into the adjective (right sub-figure). Larger circles represent a larger proportion of participants for any given proportion of regressed trials. *Metaphor-supporting* trials are shown in green, *Stance-marking* in orange, and *Literal-supporting* in purple.

The figure shows that the proportions are similar across conditions, with people regressing out of *sometimes*, on average, on 16% of *Metaphor-supporting* trials, 14% of *Stance-marking* trials, and 13% of *Literal-supporting* trials.

Given the second pass reading time differences attested at the adjective, we extend the regression analysis to regressions into the adjective region, shown on the right-hand side of Figure 4. This combination gives us an indication of not just whether people spent longer re-reading the adjective in the *Literal-supporting* condition, but also whether people were more likely to return to this point at all during the trial, compared to the other conditions. In order to quantify uncertainty over the results reported below, we fitted a Bayesian hierarchical model predicting the probability of the adjective being regressed into on each experimental trial as a function of the experimental conditions, with the condition *Literal-supporting* dummy-coded as the reference level. The model estimated the log odds of each observation being categorized as either regressed into or not. Below we report probabilities

converted from the (predicted) log odds (β) as well as 95% credibility intervals around those estimated probabilities (95% CrI [lower bound, upper bound]). We plot the proportions of regressed trials as predicted by the model atop the empirical means shown on the right-hand side of Figure 4. The model included the maximal random effect structure justified by the design, which is random slopes and intercepts for both participants and items. We model group-level variation within the items nesting them within the animal nouns. The model, fitted using the *R* package `brms` (Bürkner, 2017), had the following form, shown in `brms` syntax:

$$\begin{aligned} \text{regression} \sim \text{condition} + \\ (1 + \text{condition} \mid \text{participant}) + \\ (1 + \text{condition} \mid \text{animal/ item}) \end{aligned}$$

The figure shows that, on average, people regress into the adjective on 35% of *Metaphor-supporting* trials ($\beta = 0.29$, 95% CrI [0.15, 0.43]), 29% of *Stance-marking* trials ($\beta = 0.22$, 95% CrI [0.11, 0.34]), and 42% of *Literal-supporting* trials ($\beta = 0.39$, 95% CrI [0.26, 0.52]). In other words, people are more likely to re-read the adjective on *Literal-supporting* trials compared to both *Metaphor-supporting* and *Stance-marking* trials, with an estimated median difference of 9 percentage points (95% CrI [-0.24, 0.06]) and 16 percentage points (95% CrI [-0.30, -0.03]), respectively. Overall, this suggests that readers have more difficulty integrating the non-literal sentence continuation when the word preceding the metaphoric noun is supposed to support a literal interpretation compared to when it is supposed to support a metaphoric interpretation and to when *real* is modifying the noun.

3.3.4. Interim summary

The results of Exp. 2 show that people do not read the critical word *sometimes* faster in *Stance-marking* constructions compared to *Literal-supporting* constructions, as was found to be the case in the self-paced reading experiment (Exp. 1). Instead, the eye-tracking data show that people spend overall less time on the adjective in *Stance-marking* constructions compared to *Literal-supporting* constructions, as evidenced in both second pass and total reading times. Additionally, while regressing out of *sometimes* at similar rates across conditions, there is strong evidence that people are more likely to regress into the adjective on *Literal-supporting* trials compared to *Stance-marking* trials. All in all, this suggests less difficulty integrating the stance

adjective *real* into an unfolding non-literal discourse model compared to its *Literal-supporting* counterparts like *furry*.

An important consideration here is the differences in the methods used. In self-paced reading, readers are presented with each word individually which may encourage a more incremental style of processing. At *sometimes* they are therefore presented with a potential anomaly (based on how the sentence has been parsed) and have no choice but to try to resolve any difficulty at that point. In eye-tracking, however, the sentence is presented all at once, and readers have the benefit both of parafoveal preview of the upcoming word, and also the ability to regress and revisit prior words to help resolve any potential difficulty. This may mean that an overall more global approach is adopted, which is what our results seem to reflect (more regressions back to the adjective, but not necessarily from *sometimes*). Crucially, despite differences in exactly when and how this shows up, Exp. 1 and 2 both show a difference for the *Literal-supporting* condition compared to the other two conditions, which do not differ from each other.

Another potential consideration is the difference in word and bigram frequency for the items in Exp. 1 and 2. Crucially, despite demonstrating the longest second pass reading times, bigram frequency for literal items (e.g. *furry weasel*) was actually higher than either metaphor or stance-marking conditions (based on frequency counts from NOW, BNC and COCA), hence this is unlikely to be driving the pattern observed here. Adjective frequency, however, was lowest for the literal items, although lexical effects such as word frequency would be expected to show up in early measures (Staub & Rayner, 2007), hence again this is unlikely to explain the patterns observed here. Additional analyses available in the supplementary materials included both bigram and adjective frequency in the model for second pass reading time, and this did not change the overall pattern, although it did increase uncertainty around the effects somewhat.

4. Experiment 3 - Eye-tracking while reading (extended design)

In Experiment 3, we aimed to extend the results of Exp. 2 to further explore how the intensifying adjective *real* is interpreted in different contexts. We again utilized eye-tracking while reading to provide a sensitive way of measuring incremental reading, and manipulated the conditions to include a literally-biasing context as well as a metaphorically-biasing one. Ethics

approval was obtained from the Ethics Review Board at the University of Birmingham.

4.1. Method

4.1.1. Participants

50 monolingual speakers of English, none of whom reported dyslexia or any other reading conditions, were recruited from the student population of the Department of Linguistics and Communication of the University of Birmingham, United Kingdom. Participants received course credit in exchange for their participation in the experiment, as part of a lab session incorporating a range of studies. Data collection was conducted in a dedicated laboratory facility at the Department of Linguistics and Communication and all collected data was stored on university servers.

4.1.2. Materials and design

In Exp. 3 we extended the original design of Exp. 1 and 2 so that stance and literality would be fully crossed in a 2x2 design. Crucially, we modified the sentences in the *Literal-supporting* condition so that they would be clearly literal, and we introduced an additional condition, *Stance-marking (literal)*, where the stance marker *real* also appeared in clearly literal statements. Altogether, in our extended design, each animal noun appeared in one of four experimental conditions, as in the quadruplet below:

Metaphor-supporting (metaphor) They are *sneaky* weasels and often try to screw other people over.

Stance-marking (metaphor) They are *real* weasels and often try to screw other people over.

Stance-marking (literal) They are *real* weasels and they often live in burrows in the woods.

Literal-supporting (literal) They are *furry* weasels and they often live in burrows in the woods.

Additionally, we modified the items in the following ways: due to potential biases associated with the gendered pronouns (which might be interpreted as a cue toward a human subject), we replaced the singular pronouns *he* and *she* with *they* across all items and nouns all appeared in plural forms; we

extended the number of unique items from 15 to 44, to provide more item power; we removed any lexical phrases such as *spring chicken* or *blind bat*, which may be particularly high in frequency but which may also have lexicalized meanings of their own; and we further revised the critical adjectives so that they denote attributes which are both more and less conventionalized with the respective animal nouns they were paired with. We describe the procedure used to norm the conventionality of the different metaphor vehicles in the supplementary materials, but in brief, we created a norming task where a separate set of participants were asked to provide attributes in response to a set of animal nouns used to refer to people (all animals from Exp. 1 and 2, plus an expanded set). Based on responses, we chose the most common attribute, which ranged from very high agreement (reflecting high conventionality, e.g. *chicken* where 80% of respondents gave an attribute related to 'scared') to very low agreement (e.g. *cat*, where no attribute was provided by more than 10% or people). Metaphorical adjectives were chosen based on this to reflect the properties given, ranging in conventionality scores, and literal adjectives were chosen to match these as closely as possible for length and frequency (see Table 2), although bigram frequency did differ between conditions. Within this, some repetition was unavoidable, but we made sure that no adjective was repeated more than three times across all items. The items were counterbalanced across four presentation lists so that each animal term appeared in each of the four critical conditions in different lists. Adjectives were not repeated within a list, and the overall composition of lists in terms of spread of conventionality ratings, adjective and noun length, etc. were balanced as far as possible. Note that by definition items with 'real' (stance-literal and stance-metaphorical), which is the crucial comparison in Exp. 3, were identical in both adjective and bigram frequency. 60 filler items of comparable structure and sentence length were added to make a total of 104 items per list. The items were finally put into an eye-tracking experiment with the same parameters as in Exp. 2.

	Metaphorical adjective			*real*	Literal adjective			Animal		
	Length (chars)	Freq (Zipf)	Bigram (Zipf)	Bigram (Zipf)	Length (chars)	Freq (Zipf)	Bigram (Zipf)	Length (chars)	Freq (Zipf)	Dominance (0/20)
Mean	5.6	3.8	0.1	0.3	5.8	4.0	0.8	5.0	3.7	9.2
SD	1.7	0.7	0.9	0.6	1.7	1.0	1.2	1.7	0.6	4.5
Range	3-10	2.5-5.3	-1.4-2.1	-0.8-1.3	3-10	1.8-5.3	-1.4-2.7	3-9	2.5-4.6	2-19

Table 2: Statistical properties of the critical adjective-noun pairs in Exp. 3. All frequencies are taken from the NOW corpus and converted to Zipf frequencies (as per van Heuven et al., 2014).

The items were counterbalanced across four presentation lists so that each animal term appeared in each of the four critical conditions in different lists. Adjectives were not repeated within a list, and the overall composition of lists in terms of spread of conventionality ratings, adjective and noun length, etc. were balanced as far as possible. 60 filler items of comparable structure and sentence length were added to make a total of 104 items per list. The items were finally put into an eye-tracking experiment with the same parameters as in Exp. 2.

4.1.3. Procedure

The procedure was the same as in Exp. 2. A total of 39 items (including all conditions and filler items, accounting for just over 1/3 of the total) included comprehension questions.

4.2. Hypotheses

The hypothesis entertained here is the same as in Exp. 1 and 2, namely that stance adjectives act as strong cues to non-literality in the processing of metaphoric statements. Given the revised experiment design, we focus on the comparison between the *Stance-marking (literal)* and *Stance-marking (metaphor)* items, which differ in terms of the sentence continuation after the critical noun. In particular, we are interested in comparing the two *Stance-marking* conditions at the adjective and noun regions. We expect that *Stance-marking (metaphor)* items will be processed more easily than *Stance-marking (literal)* items, given that in the former the sentence continuation is non-literal while in the former it is literal. This can translate into shorter second pass reading times at the adjective and/ or noun as well as lower rates of regression into those same regions in *Stance-marking (metaphor)* items, as evidenced by the results from Exp. 2. In other words, if *real weasel* is understood by default as a stance-marker / figurative description,

then the literal continuation (where *real* = a literal weasel) will require more regressions and re-reading to reconfigure the literal understanding.

4.3. Results

The data and analysis scripts are available for inspection under `https://osf.io/u6f98/`. Data was pre-processed using Eyelink DataViewer software (version 4.4.1).

4.3.1. Data cleaning

The data was cleaned in the same way as described for Exp. 2. Usable data from 42 participants was analyzed. 4.5% of usable data was removed, leaving 1765 trials for analysis. According to our pre-registered plans, any participant who did not reach an accuracy of 80% on the comprehension questions would be removed. All participants exceeded this threshold (mean = 97.0%, SD = 17.7).

4.3.2. Reading times

First, we report the mean reading times across sentence regions, as shown in Figure 5 below. All relevant results are summarized in Table 3 below.

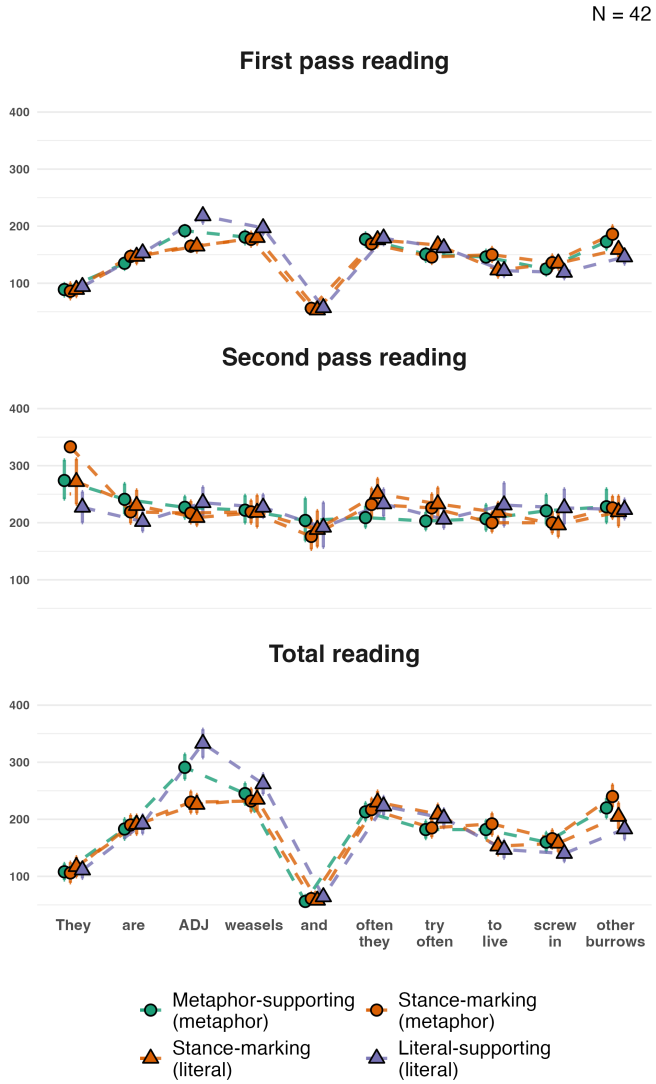


Figure 5: Mean reading times (y-axis) at each sentence region (x-axis). The top panel shows first pass reading times, the middle one second pass reading times, and the bottom one total reading times. *Metaphor-supporting* items are shown in green, *Stance-marking* in orange, and *Literal-supporting* in purple. The error bars represent 95% bootstrapped confidence intervals. Note that second pass reading times do not include any zero values.

		Met. (met.)	Stance (met.)	Stance (lit.)	Lit. (lit.)
adjective	First pass	192 (110)	165 (106)	165 (111)	218 (152)
	Regression path	312 (255)	258 (258)	257 (273)	313 (234)
	Second pass⋆	75 (126)	49 (102)	48 (95)	85 (150)
	Second pass*	227 (117)	217 (101)	209 (77)	235 (166)
	Total	291 (233)	230 (202)	226 (185)	333 (280)
	Reg. IN	0.29 (0.45)	0.25 (0.43)	0.23 (0.42)	0.32 (0.47)
	Reg. OUT	0.27 (0.44)	0.19 (0.40)	0.22 (0.42)	0.26 (0.44)
noun	First pass	181 (128)	177 (126)	180 (131)	197 (128)
	Regression path	281 (278)	271 (321)	264 (238)	323 (300)
	Second pass⋆	52 (112)	43 (97)	46 (109)	56 (112)
	Second pass*	222 (126)	219 (96)	218 (139)	227 (108)
	Total	245 (192)	232 (207)	235 (203)	262 (196)
	Reg. IN	0.14 (0.34)	0.17 (0.38)	0.14 (0.35)	0.14 (0.35)
	Reg. OUT	0.21 (0.41)	0.18 (0.38)	0.19 (0.39)	0.26 (0.44)

Table 3: Reading measures at the adjective and noun regions. Reading times are shown in milliseconds, regressions in percentages (with standard deviations within parentheses). ⋆ = Second pass reading times including zeroes (i.e. where no re-reading took place); * = Second pass reading times not including zero values (no re-reads).

The figure shows that, at the adjective, *Metaphor-supporting (metaphor)*, *Stance-marking (metaphor)*, and *Stance-marking (literal)* items are all read faster than *Literal-supporting (literal)* items in first pass reading times (differences = 26, 53, and 53 ms faster, respectively). The results also show that *Stance-marking (literal)* items are read similarly fast to *Stance-marking (metaphor)* items. Crucially, differences between conditions are negligible for second pass reading times, unlike what was predicted by our baseline expectation and by the hypothesis of stance adjectives as strong cues to non-literality. At the noun, the differences between all conditions are negligible for all reading measures, unlike what was predicted by our baseline expectation and by the hypothesis of stance adjectives as strong cues to non-literality. Table 4 confirms that there are no noticeable differences between *Stance-marking* conditions for any measure at either the adjective or the noun.

As in Exp. 2, we focus on the second pass reading times at the adjective (ADJ) as a plausible marker of processing difficulty (reanalysis as a result of misattribution during initial reading of *real*). In order to quantify uncertainty over the results reported below, we fitted a Bayesian hierarchical model predicting log-transformed reading times at the adjective region as a function of the experimental conditions, with the condition *Literal-supporting (literal)* dummy-coded as the reference level. Given that readers may not necessarily

read a word a second time after first pass, we excluded trials on which second pass reading times were 0 (71% of the data at the adjective). In order to account for the light right skew in the response distribution, we fitted a model using a skewed normal distribution. The model included random slopes and intercepts for both participants and items. We model group-level variation within the items nesting them within the animal nouns. The model, fitted using the *R* package `brms` (Bürkner, 2017), had the following form, shown in `brms` syntax:

$$\begin{aligned} \log(\text{reading time}) \sim \text{condition} + \\ (1 + \text{condition} \mid \text{participant}) + \\ (1 + \text{condition} \mid \text{animal/ item}) \end{aligned}$$

Visual inspection of the results in Fig. 5 suggests that, at the adjective region, differences between conditions are negligible in second pass reading times. These results finds strong quantitative support in the respective statistical model (see Table 4 below). In other words, our prediction that *real* would be understood as stance-marking by default and hence lead to more disruption and need for reanalysis in the *Stance-marking (literal)* condition was not supported. In fact, minimal differences across all conditions suggest that no reading of the adjective led to any more disruption than any other.

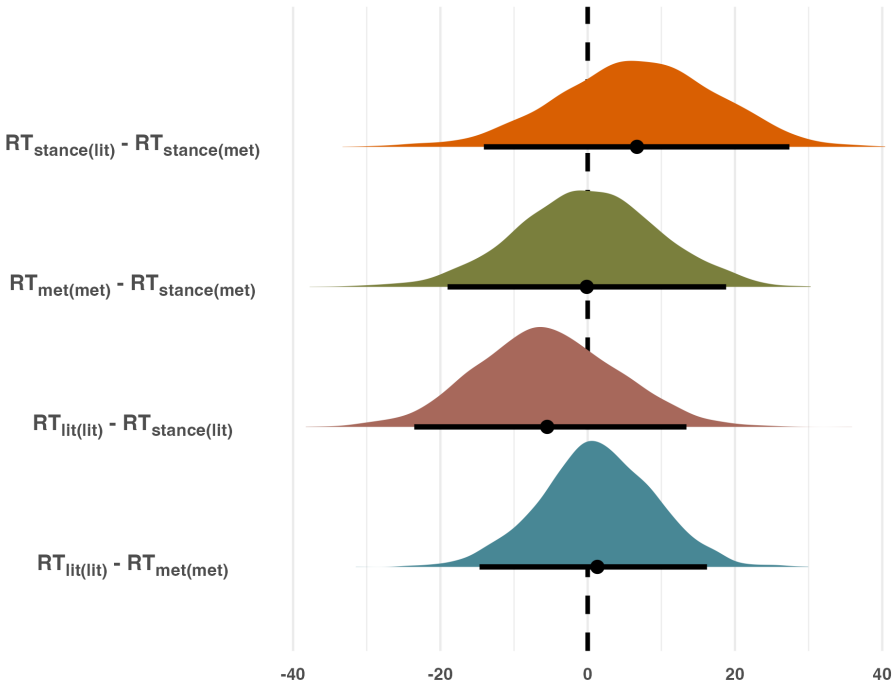


Figure 6: Differences in estimated second pass reading times at the adjective (x-axis). The difference between *Literal-supporting (literal)* and *Metaphor-supporting (metaphor)* items ($RT_{lit(lit)}$ - $RT_{met(met)}$) is shown in blue, the one between *Literal-supporting (literal)* and *Stance-marking (literal)* items ($RT_{lit(lit)}$ - $RT_{stance(lit)}$) in red, the one between *Metaphor-supporting (metaphor)* and *Stance-marking (metaphor)* items ($RT_{met(met)}$ - $RT_{stance(met)}$) in green, and the one between *Stance-marking (literal)* and *Stance-marking (metaphor)* items ($RT_{stance(lit)}$ - $RT_{stance(met)}$) in orange. The black circles represent the mean estimated differences and the horizontal bars represent the respective 95% credibility intervals.

Measure	Hypothesis	β	95%-CrI	P($\beta > 0$)
Second pass	Lit. (lit.) > Met. (met.)	1.3	[-13.3, 17.6]	0.57
Second pass	Lit. (lit.) > Stance (lit.)	-5.5	[-23.5, 13.4]	0.28
Second pass	Met. (met.) > Stance (met.)	-0.1	[-19.0, 18.8]	0.50
Second pass	Stance (lit.) > Stance (met.)	6.7	[-14.1, 27.4]	0.74

Table 4: Regression coefficients.

4.3.3. Regression rates

Figure 7 below shows the mean proportion of regressions into the adjective.

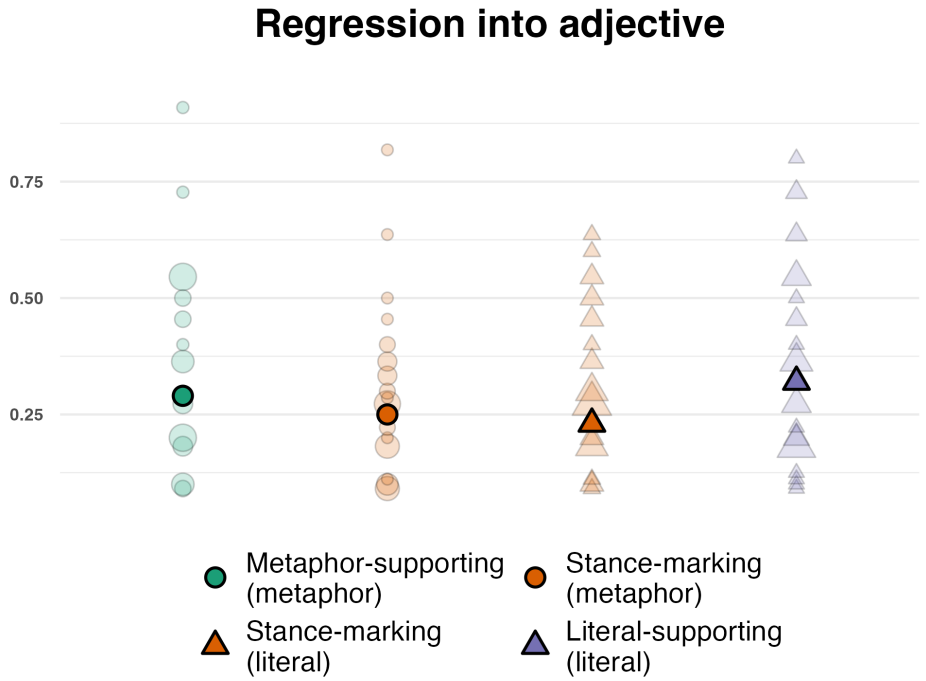


Figure 7: Mean proportions of regressed trials (y-axis). Mean proportions of regressed trials (fully colored shapes), overlaid to participant means (shaded shapes with varying sizes). Values closer to 1 on the y-axis represent a larger proportion of trials on which participants regressed into the adjective. Larger shapes represent a larger proportion of participants for any given proportion of regressed trials. *Metaphor-supporting (metaphor)* items are shown in green, *Stance-marking (metaphor)* in orange (circles), *Stance-marking (literal)* in orange (triangles), and *Literal-supporting (literal)* in purple.

The figure shows that the proportions are similar in the *Stance-marking (metaphoric)* and *Stance-marking (literal)* conditions (25 and 23%, respectively), unlike what was predicted by the hypothesis of stance adjectives as strong cues to non-literality.

4.3.4. Interim summary

The results of Exp. 3 show that people read neither the stance adjective nor the noun more slowly in *Stance-marking (literal)* constructions compared

to *Stance-marking (metaphor)* ones, unlike what was predicted. By the same token, people do not regress back into the adjective more often in either *Stance-marking* condition, also unlike our predictions.

In addition to looking at the reading measures alone, we explored how these results are affected by the degree of conventionality of different metaphoric vehicles, looking at the correlation between reading times and regression rates and the metaphoricity scores mentioned in Section 4.1.2. The rationale of this exploration was based on the assumption that more conventionalized vehicles should serve as stronger cues to non-literality than less conventionalized ones, in line with previous work showing a substantial role of conventionalization in metaphor processing (e.g. Cardillo et al., 2012; Bowdle and Gentner, 2005). In other words, more conventionalized animal terms (e.g. *weasel*, where a specific property is generally agreed upon) should be easier to integrate figuratively and more difficult to integrate literally, whereas a less conventionalized term (e.g. *cat*, where multiple interpretations may be possible) should be easier to integrate literally and harder to integrate figuratively. The results of the exploratory analyses show, by and large, this pattern for literal items, whereby as conventionality increases, both reading times and regression rates increase. However, this was also found to be the case for metaphorical items, which is contrary to what was expected, as higher conventionality was expected to correlate with shorter reading times and lower regression rates. Overall, conventionality had limited impact on reading patterns for metaphorical items. Due to the length of these analyses, they are reported in detail in the supplementary materials (along with details of how conventionality ratings for individual items were obtained).

5. Discussion

In the present study, we set out to test whether adjectives with a conventionalized meaning of intensification (i.e. *real*) act as predictive cues to non-literality in the online processing of language. In three experiments we compared the processing of metaphorical and literal statements, and compared lexical adjectives with *real*, which can have both a conceptual (real = genuine) and a stance-marking (real = intensifier) role.

Exp. 1 and 2 supported the prediction that *real* is interpreted by default as an intensifying adjective in that items with *real* were read comparably to more clearly metaphorically-biasing items like *devious weasel*. In contrast when readers arrived at a part of the sentence that clearly marked non-

literality ("sometimes"), literally-biasing items showed disruption. This was evident in longer reading times for "sometimes" itself in Exp. 1, whereas in Exp. 2 readers more likely to return to re-read the adjective as a way of resolving the encountered incongruency.

Exp. 3 extended the design by adding a fourth condition to allow us to compare *real* in both metaphorical and literal sentences. If the results of Exp. 1 and 2 suggest that *real* is by default interpreted as stance-marking, and hence more likely to cue a non-literal meaning, then the stance-marking (literal) condition (e.g. *They are real weasels and often live in burrows in the woods*) should have been problematic, with evidence of recovery through more regressions and longer re-reading times once the incongruent context was encountered. This was not supported, with no differences between the two stance-marking conditions for any eye-tracking measure. This suggests that both meanings were considered, as we discuss further below.

Overall, the results suggest that intensifying adjectives can act as pragmatic cues in the incremental interpretation of language. In particular, the results highlight how such markers of stance serve as *predictive* cues to non-literal meaning, such that in the processing of non-literal, metaphoric statements they cue a *potential* non-literal interpretation despite their lexical semantics related to truth or reality. This is generally in line with the hypothesis that, given its function as a marker of subjective stance, *real* acts as a strong cue to non-literality when used in non-literal utterances. Underlying this hypothesis was the assumption that, as a cognitive process, interpretation involves dynamically weighing the cues available in a communicative signal, such that at any given moment in processing the most likely outcome of the interpretive process consists of the combined weight of all cues processed thus far (Rohde and Kurumada, 2018). Key to the interpretive process is the fact that the pragmatic interpretation is cued by the meaning of intensification conventionally associated with the adjectival form *real*, which renders a non-literal interpretation more likely than its literal counterpart.

From a psycholinguistic perspective, there are various cognitive factors known to affect the ways in which comprehenders access the most likely meaning representations in their mental lexicons. The literature on lexical ambiguity resolution suggests that multiple meanings are activated, even in biasing contexts. Indeed, while a biasing preceding context can help to quickly home in on the more appropriate interpretation, the data generally agrees on some form of "exhaustive" access, whereby multiple meanings are activated on encounter (e.g. Binder and Rayner, 1998; Kambe et al.,

2001; Dopkins et al., 1992; Swinney, 1979; Duffy et al., 1988). In our study, as discussed in the introduction, this process can be grounded in reasoning about conventional implicatures (Potts, 2007, 2003, 2012), such that the conventionalized meaning of *real* as an intensifier supports the metaphoric interpretation of *He is a real weasel*. This provides a parsimonious explanation for the pragmatic interpretation of *real*: its intensificational meaning provides a functionally relevant cue in context, despite the potentially conflicting conceptual meaning which is activated alongside it. An alternative heuristic which might be said to account for the interpretation at hand is one which assigns *real* a role as a procedural cue. In line with this view, certain, if not all, linguistic items provide cues to interpretation insofar as they encode instructions as to how to process the relevant proposition(s) in an utterance (Blakemore, 1987, 2002), either independently from or in addition to contributing to the truth-conditions of those same propositions (Carston, 2016). In the current case, one could thus assume a procedural role for *real*, such that it cues how to interpret the unfolding proposition *He is a weasel sometimes during games*.

An alternative account altogether would be one grounded in reasoning about the (Gricean) quality of the utterance: saying that someone is a *real weasel* is blatantly false, especially when using a gendered singular pronoun like *he* or *she*, and so by virtue of the proposition being explicitly not true it likely means that the utterance is meant non-literally. However, while this might explain the interpretation of a construction where the word preceding the noun straightforwardly supports the metaphoric meaning of the noun given its conceptual meaning, like *devious weasel*, it does not explain how the conceptual meaning of 'reality' of *real* would override its intensifying meaning, at least not without additional assumptions. The data from Exp. 3 does not support a view in which either meaning is more salient than the other, which renders a quality account less parsimonious than a conventional implicature and a procedural one.

It is important to note that in Exp. 1 and 2 the gendered pronouns *he* and *she* may contribute to constructions like *He is a real weasel* cueing a metaphorical meaning. Similarly, the results of Exp. 3, where the gendered pronouns were replaced with *they*, while still providing evidence that *real weasel* cues a metaphorical meaning also suggest that it cues a literal one as well. In other words, both the intensificational and the conceptual meaning of *real* are activated and maintained in processing, neither seemingly having a preference over the other, even when further cues to non-literality enter

the frame downstream from the metaphoric noun, as in the *Stance-marking (metaphor)* condition in Exp. 3. We note here that our interpretation of *real* as cuing a non-literal meaning is not unequivocally indicated by our results, and we have no way of knowing that readers actually did interpret these conditions metaphorically (just as we have no way of knowing that phrases like *devious weasel* were definitely understood as metaphors, and *furry weasel* were definitely understood as literal). We can, however, say that *real weasel* was not seen as anomalous in the same way as the literal (*furry weasel*) condition (in Exp. 1 and 2), which is likely to have been the case if *real* was interpreted at its lexical face value. In Exp. 3, *real* was read and re-read almost identically, regardless of how the sentence continued, hence neither reading was seen as anomalous (relative to the other). To better understand how both the literal and the stance-marking conditions were actually understood, the addition of more probing comprehension questions may be illuminating.

Finally, it is also important to note that distributional properties of the items differed between conditions, and whilst this was controlled in Exp. 3 for aspects such as individual word frequency, there were still differences between conditions for bigram frequencies. There was also substantial repetition in the stance-marking conditions, hence readers will have seen items with *real* multiple times, compared to other conditions, which may have led to confusion in how this should be interpreted. In future studies, more variation in items (e.g. other markers that serve similar functions to *real*) may help overcome this, as may a between-subjects design where participants see *real* in only literal or metaphorical uses. While these statistical regularities may have contributed to the pattern of results observed, we again stress that in Exp. 3 literal-supporting adjectives were not re-read more slowly than metaphor-supporting ones, nor was *real* re-read more slowly in either literal or metaphorical sentences. That is to say that, despite differences in surface regularities, comprehenders processed all modifiers similarly, as indexed by second pass reading times and regression rates.

The current results add to the body of evidence suggesting that the relative weight of an adjective – or any other linguistic item, for that matter – varies depending on its functional relevance for interpretation in context (Gardner et al., 2021; Rubio-Fernandez and Jara-Ettinger, 2020; Tourtouri et al., 2021). Crucially in the case of an adjective with a conventionalized meaning of intensification, such functional relevance seems to be only weakly constraining, probably given that the adjective contributes both a lexically

encoded conceptual meaning *and* an intensificational meaning. In fact, as discussed in the introduction, different lexical items impart more or less of their conceptual meaning to their conventionalized meaning as intensifiers, such that in the case of *real* one cannot straightforwardly distinguish one meaning from the other without further cues, either intrasentential or from the larger extrasentential discourse context.

All in all, the present study sheds light on the potential mechanisms underlying the interpretation of stance markers such as the intensifying adjective *real*. On the face of it, interpreting an utterance like *He is a real weasel* involves activating both the straightforward conceptual meaning of *real* as a truth marker item and its conventionalized meaning as a marker of epistemic stance. Deriving therefrom the most likely interpretation in context can be said to involve drawing an inference on the basis of the adjective's conventionalized meaning – although its conceptual meaning remains activated alongside it without hindering the interpretive process. Ultimately, these results speak to the activation of secondary meaning and its impact on the pragmatic processing of language, supporting a structured underspecification view (e.g. Ortega-Andrés and Vicente, 2019, see also Carston, 2021; Li, 2024) in which epistemic stance markers contribute both a conceptual and a conventionalized stance meaning. That is, upon encounter, *real* activates separate but overlapping meanings, with the language processor maintaining both until sufficient additional evidence has been encountered to rule one out.

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